Assignment-1

Part I: Short Answer Questions

1. A DBMS stores data in an organized (structured) form with the usage of tables and it ensures data security, with easy access via the usage of queries. A traditional file system, however, stores the data as raw files (without structure), no built-in mechanism for relationships, queries etc. and no security of data.
   A good usage of DBMS would be to store the records (name, roll number, class, address) for all the students in a school. This would also be much faster than a traditional file system, especially if the number of students is large.

2. In ER diagram, a strong entity (e.g. Roll No.) is represented by a rectangle, whereas a weak entity (e.g. Name) is represented by a double rectangle.
   The primary key of weak entity is formed by its relationship with the strong entity called the identifying relationship (represented by a double diamond).

3. It should be a multi-valued attribute as a student may not necessarily have just one phone number (simple) and a phone number cannot be broken into multiple meaningful subparts (composite).

4. A super key is a set of attributes that can uniquely identify a row in the database. A candidate key is a smallest possible super key with no redundant attributes. For the given example, {roll_no} and {email} are the candidate keys. {roll_no, name} and {email, branch} are super keys which are not candidate keys.

5. The Referential Integrity Constraint is violated here as the Student and Enrollment tables will no longer be consistent anymore due to the additional student_id in enrollment table. The consistency ensures that every enrollment entry corresponds to a valid student in the database. The insert option should be made invalid so that such records cannot be entered to ensure consistency.

6. A table showing the relationship between the entities has to be created as one of the keys for an entry might correspond to NULL due to no mapping (for example, a course might not have any students). The primary key of this table would be formed by the combination of the primary keys of the participating entities.

Part II: Long Answer Question

Part A

There are 4 entities here –

1. Customer
2. Restaurant

3. Menu Item
4. Order

The key attribute can be any unique ID number assigned to them.

1. Customer-Order Relation is 1:N as a customer can place multiple orders but any order is traced back only to a specific customer.
2. Restaurant-Order Relation is 1:N as each order is given to exactly one restaurant but a restaurant can take multiple orders.
3. Order-Menu Item Relation is M:N as an order can have multiple items and the same item can appear in multiple orders.
4. Restaurant- Menu Item Relation is 1:N as a restaurant can offer multiple items but each item belongs to one restaurant.

Part B

For customers, a multivalued attribute is Phone Number as a customer can have multiple phone numbers for different locations, and store them all in the app. A composite attribute is Address as it contains multiple sub components like city, state, pin code etc.

Part C

Order&MenuItem (Order ID, MenuItem ID, Qty)

The primary key is (Order ID, MenuItem ID), a combination of the unique key attributes of the entities. Here OrderID and MenuItemID are the foreign keys, referencing Order and Menu Item respectively. Quantity is an additional attribute here.

Part III